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import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

from sklearn.preprocessing import StandardScaler
from sklearn.cluster import KMeans
from sklearn.metrics import silhouette_score

df = pd.read_csv("/content/CC GENERAL.csv")

print("Missing Values:\n", df.isnull().sum())

df = df.drop(columns=['CUST_ID'], errors='ignore')
df.fillna(df.mean(), inplace=True)

scaler = StandardScaler()
scaled_data = scaler.fit_transform(df)

wcss = []
for i in range(1, 11):
    kmeans = KMeans(n_clusters=i, random_state=42, n_init=10)
    kmeans.fit(scaled_data)
    wcss.append(kmeans.inertia_)

plt.figure()
plt.plot(range(1, 11), wcss, marker='o')
plt.title("Elbow Method")
plt.xlabel("Number of Clusters")
plt.ylabel("WCSS")
plt.show()

print("\nSilhouette Scores:")
for i in range(2, 11):
    kmeans = KMeans(n_clusters=i, random_state=42, n_init=10)
    labels = kmeans.fit_predict(scaled_data)
    score = silhouette_score(scaled_data, labels)
    print(f"Clusters: {i}, Score: {score}")

optimal_clusters = 4
kmeans = KMeans(n_clusters=optimal_clusters, random_state=42,
n_init=10)
df['Cluster'] = kmeans.fit_predict(scaled_data)

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cluster_summary = df.groupby('Cluster').mean()
print("\nCluster Summary:\n", cluster_summary)

df.to_csv("segmented_customers.csv", index=False)

print("\n Clustering Completed! File saved as
'segmented_customers.csv'")

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Missing Values:

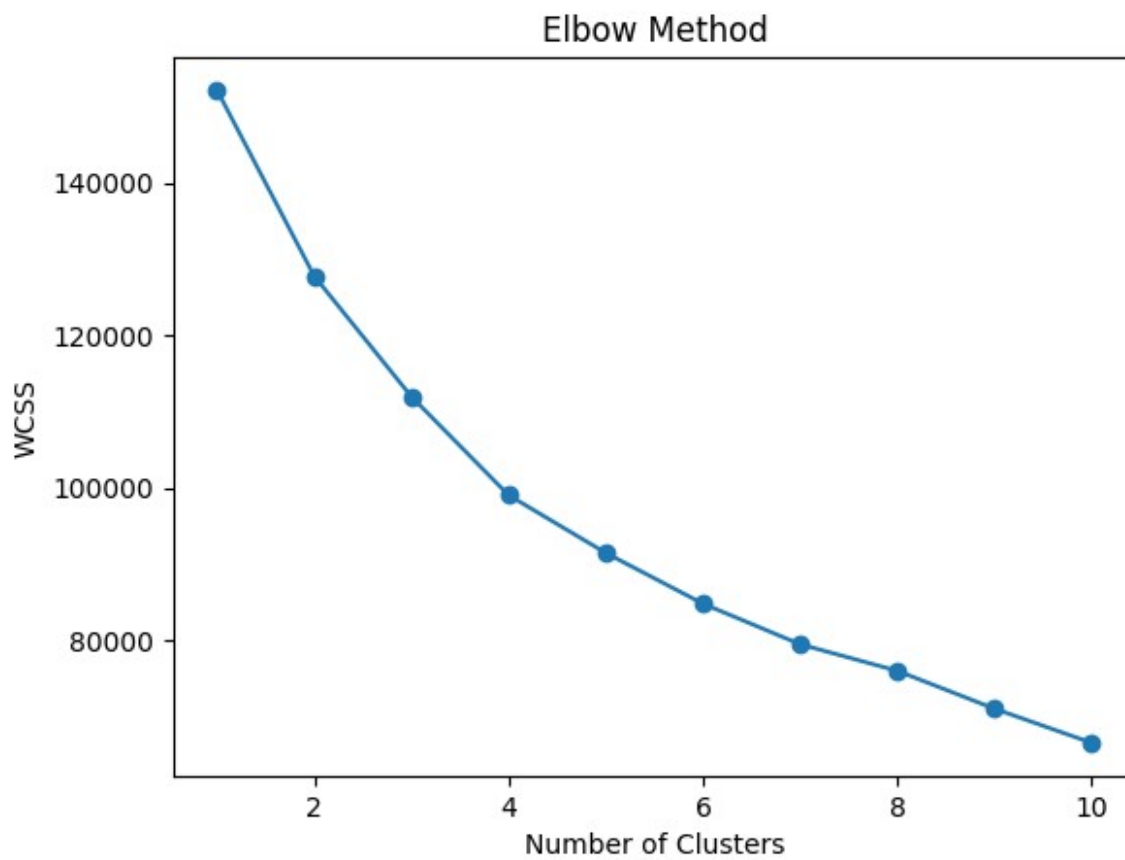
```

CUST_ID	0
BALANCE	0
BALANCE_FREQUENCY	0
PURCHASES	0
ONEOFF_PURCHASES	0
INSTALLMENTS_PURCHASES	0
CASH_ADVANCE	0
PURCHASES_FREQUENCY	0
ONEOFF_PURCHASES_FREQUENCY	0
PURCHASES_INSTALLMENTS_FREQUENCY	0
CASH_ADVANCE_FREQUENCY	0
CASH_ADVANCE_TRX	0
PURCHASES_TRX	0
CREDIT_LIMIT	1
PAYMENTS	0
MINIMUM_PAYMENTS	313
PRC_FULL_PAYMENT	0
TENURE	0

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dtype: int64

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Silhouette Scores:

Clusters: 2, Score: 0.21003302619400505
Clusters: 3, Score: 0.25055570796307736
Clusters: 4, Score: 0.1976074492720698
Clusters: 5, Score: 0.1931688590154638
Clusters: 6, Score: 0.20263931739848778
Clusters: 7, Score: 0.21503808606365246
Clusters: 8, Score: 0.20809196271641647
Clusters: 9, Score: 0.2149116403835865
Clusters: 10, Score: 0.2205181120683996

Cluster Summary:

	BALANCE	BALANCE_FREQUENCY	PURCHASES	ONEOFF_PURCHASES \
Cluster				
0	894.907458	0.934734	1236.178934	593.974874
1	3551.153761	0.986879	7681.620098	5095.878826
2	4602.462714	0.968415	501.896219	320.373681

3	1011.751528	0.789871	269.973466	209.853863
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	INSTALLMENTS_PURCHASES	CASH_ADVANCE	PURCHASES_FREQUENCY	\
Cluster				
0	642.478274	210.570626	0.885165	
1	2587.208264	653.638891	0.946418	
2	181.607404	4520.724309	0.287731	
3	60.386625	595.759339	0.170146	

	ONEOFF_PURCHASES_FREQUENCY	PURCHASES_INSTALLMENTS_FREQUENCY	\
Cluster			
0	0.297070	0.711842	
1	0.739031	0.788060	
2	0.138934	0.185516	
3	0.086281	0.080578	

	CASH_ADVANCE_FREQUENCY	CASH_ADVANCE_TRX	PURCHASES_TRX	\
Cluster				
0	0.042573	0.790021	22.091773	
1	0.071290	2.085575	89.359413	
2	0.484526	14.284641	7.661102	
3	0.114833	2.125503	2.903421	

	CREDIT_LIMIT	PAYMENTS	MINIMUM_PAYMENTS	PRC_FULL_PAYMENT	\
Cluster					
0	4213.207678	1332.194205	650.167072	0.269258	
1	9696.943765	7288.739497	1976.815179	0.286707	
2	7546.957050	3481.145990	2008.251157	0.034859	
3	3277.703165	974.505090	586.301239	0.078001	

	TENURE
Cluster	
0	11.594595
1	11.951100
2	11.387312
3	11.446429

□ Clustering Completed! File saved as 'segmented_customers.csv'